PAIDIPILLI PURUSHOTHAM

ECE Student: VLSI Design, ASIC Verification, & Embedded Systems

23pa1a04b4@vishnu.edu.in | +91-8093853830

Bhimavaram, AP | LinkedIn | GitHub | YouTube

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE) 08/2023 – 08/2027

Vishnu Institute of Technology (CGPA: 9.73/10) Bhimavaram, AP Seeking 6-month Internship (04/2027–09/2027) and looking for Full-Time roles (PPO).

Intermediate (MPC) – Aditya Junior College (Marks: 971/1000) *06/2021 – 04/2023*

TECHNICAL SKILLS

- **Hardware & HDLs:** VLSI Design, Verilog HDL, System Verilog, RISC-V, Embedded C, CDC (Clock Domain Crossing)
- **Verification & Tools:** UVM, Synopsys VCS, Design Compiler, Pt-Shell, Custom Compiler, Sentaurus TCAD, Cadence Virtuoso, OpenLane, Xilinx Vivado, LabVIEW
- **Programming & Workflow:** Python, C Programming, Linux, GitHub
- **Frameworks:** .NET Core, Angular, SignalR, IoT, FPGA Prototyping

EXPERIENCE

Integrated VLSI Design Intern (Silicon to GDS) ***[Offer]*** 05/2026 – Present

SURE Trust / ProEd *Remote*

- Completed 3 core tracks: Semiconductor Fundamentals, RTL Design with Clock Domain Crossing (CDC), and Functional Verification using SystemVerilog/UVM; currently executing Design for Testability (DFT).
- Architected multi-module digital systems in Verilog, including a serial pseudo-random bitstream generator with FSM/shift-register pattern matchers and BCD/7-segment display logic.

VLSI Engineering Trainee 08/2024 – Present

HCL *Remote*

- Directed the formulation of the SafeALU architecture, reducing switching power consumption by 25% and ranking 1st among 62 engineering candidates.
- Executed advanced Physical Design (PD) and Verification projects using the Synopsys RTL-to-GDSII toolchain, focusing on structural coverage and timing-driven synthesis.

Summer Research Intern (VLSI Signal Processing) ***[Post]*** 05/2026 – 07/2026

IIT Bhubaneswar (Dept. of ECE) *Bhubaneswar, Odisha*

- Entrusted with full ownership of high-performance BCI signal processing pipelines across 4 massive datasets (>20GB); implemented Butterworth bandpass filtering and Euclidean Alignment (EA) for spatial covariance normalization.
- Applied Riemannian Tangent Space projection and Hybrid Stacking Ensembles (SVM, LDA, RF) to optimize EEG classification frameworks, boosting baseline accuracy by 12%.

Industrial Trainee (RF & Semiconductors) ***[Certificate]*** 05/2026 – 06/2026

GAETEC (DRDO / ISRO Foundry) *Hyderabad, TS*

- Analyzed MESFET and pHEMT device architectures in the Design section using Keysight tools, focusing on Gallium Arsenide (GaAs) fabrication across high-frequency RF spectrums.
- Evaluated ATQC (Assembly, Testing & QC) processes for 100+ components: Die/Wire bonding, Shear/Pull testing, and RF characterization via VNA/PNA; assessed PIND, Acceleration, and Thermal cycles.

PROJECTS

Advanced 32-bit RISC-V Core Tapeout (RTL-to-GDSII) ***[View Repo]*** *OpenLane, Sky130 PDK*

- Spearheaded the full physical design flow (Floorplanning, CTS, Routing) for a 32-bit RISC-V core architecture.
- Validated 100% DRC/LVS cleanliness and achieved optimal timing closure, ranking in the Top 22 nationally.

Asynchronous FIFO Design & UVM Verification ***[View EDA]*** *Verilog, SystemVerilog, UVM*

- Engineered an Asynchronous FIFO in Verilog, implementing separate clock domains for read and write logic to achieve zero data loss across 10,000+ randomized test cycles.
- Built a robust verification environment using SystemVerilog (OOP, Constrained Randomization) and UVM.
- Developed UVM components (Drivers, Monitors, Scoreboards) to verify data integrity and 100% functional coverage.

Energy Efficient VLSI Architecture for ECG Detection *Verilog, Synopsys VCS/DC*

- Architected a DSNN-based ECG accelerator by replacing multipliers with shift-and-add logic; achieved 86% accuracy.
- Synthesized the design using Synopsys DC (1321.32 sq. um) and optimized for 115.4 uW low-power operation.

Semiconductor Device Simulation & Modeling (TCAD) *Sentaurus Workbench*

- Executed full-flow simulations of NMOS, PMOS, and CMOS structures using Sentaurus TCAD Workbench, extracting precise ID-VGS and ID-VDS transfer characteristics at advanced technology nodes.
- Conducted advanced comparative analysis between standard Silicon FinFETs and SiGe FinFETs to assess carrier mobility across 3 temperature variations.

AWARDS & CERTIFICATIONS

- **1st Runner-Up (Spark Tank):** Led a 4-member cross-functional team to secure an INR 30k cash prize among 50+ teams from E-Cell VITB for the AQMD project.
- **1st Place (Hustlethon 2k25):** Built the winning IoT Aquaculture Water Testing solution out of 100+ competitors.
- **Advanced RISC-V Tapeout:** Completed an intensive 10-week RTL-to-GDSII flow for a 32-bit RISC-V core, ranking top 22 nationally. *[View GitHub Repo]*
- **HAM Radio Operator:** Hold a General License; actively operate at the Vishnu Space Applications Center.
- **Certifications:** HCL VLSI Foundation | NPTEL Elite Comm. Systems | VLSI SoC Design | NPTEL Elite+ IoT